ENGLISH

PAPER CODE : 2701CJA101001250021

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2026-2027)

JEE (ADVANCED)

MT # 04

27-07-2025

JEE (Main + Advanced) : NURTURE COURSE

PAPER # 02

PHASE–I SANKALP - (I)

Time : 3 Hours. **Maximum Marks : 180**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains 30 pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.

8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**

9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.

11. Darken the bubble ◯ **COMPLETELY.**

12. The correct way of darkening a bubble is as : ●

13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.

14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

15. Take $g = 10\ m/s^2$ unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see th e last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

SPACE FOR ROUGH WORK

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

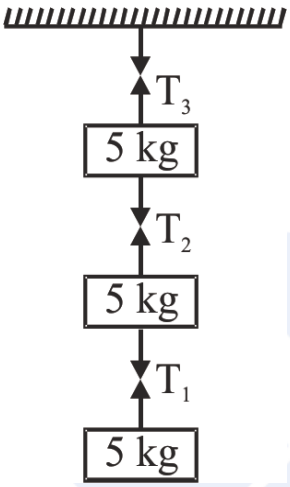
SECTION - I(i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

SPACE FOR ROUGH WORK

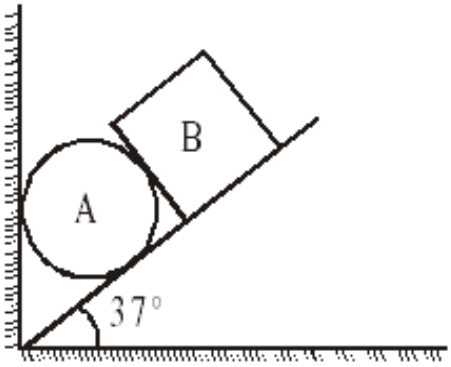
1. The value of $\frac{T_3}{T_1}$ is :-



(A) 1 (B) 2

(C) 3 (D) $\frac{3}{2}$

2. A perfect smooth sphere A of mass 2 kg is in contact with a rectangular block B of mass 4 kg and vertical wall as shown in the figure. All surfaces are smooth. Find normal reaction by vertical wall on sphere A ($g = 10m/s^2$)



(A) 20 N (B) 25 N

(C) 80 N (D) 45 N

SPACE FOR ROUGH WORK

3. An insect start moving along a rope with velocity V relative to rope. One end of rop is fix to a point A, and other end (initially at a distance L from fix point A) is pulled with speed U. After how much time later insect reaches the other end of rope (insect start motion from point A, rope stretch uniformly)



(A) $\frac{L}{U}\left(e^{U/V}-1\right)$

(B) $\frac{L}{U+V}\left(e^{U/V}-1\right)$

(C) $\frac{L}{U}\left(e^{2U/V}-1\right)$

(D) $\frac{L}{U+V}\left(e^{2U/V}-1\right)$

4. Two particles are moving with velocity v_1 and v_2 respectively which of the following is correct about relative velocity :- ($v_1, v_2 \neq 0$)

(A) Magnitude of velocity of 1 w.r.t. 2 can not be equal to $|v_1|$

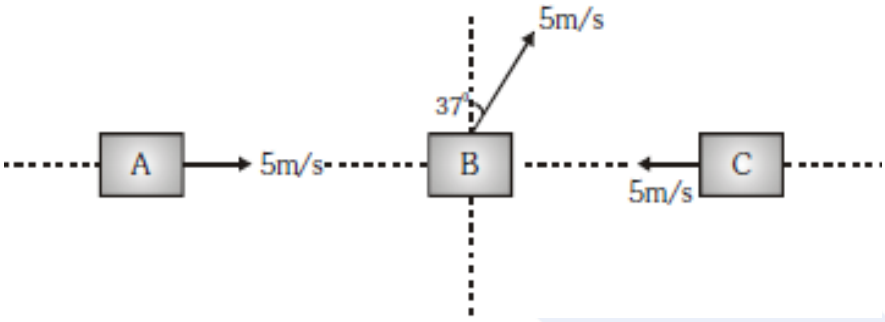
(B) Magnitude of velocity of 1 w.r.t. 2 can be smaller than $||v_1| - |v_2||$

(C) Magnitude of velocity of 1 w.r.t. 2 can not be greater than $|v_1| + |v_2|$

(D) Magnitude of velocity of 1 w.r.t. 2 can not be equal to $|v_2|$

SPACE FOR ROUGH WORK

5. Consider the motion of three bodies as shown



for an observer on B, what is the magnitude of relative velocity of A with respect to C ?

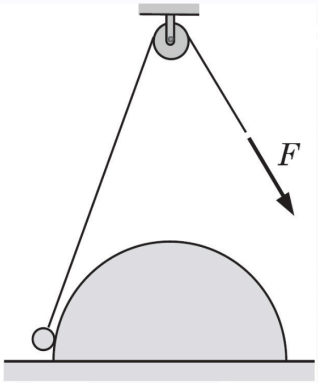
(A) 10 m/s

(B) 0

(C) $\sqrt{20}$ m/s

(D) 8 m/s

6. A small metal ball is being pulled gradually on a fixed frictionless hemisphere as shown in the figure. Radius of the ball and that of the pulley are much smaller than that of hemisphere. As the ball slides from the bottom to a position close to the top of the hemisphere, how do the magnitudes of pulling force F and normal force R between the ball and the hemisphere change?(consider configuration in which rope is not touching sphere)



(A) F increases and R decreases

(B) R increases and F decreases

(C) F decreases and R remains constant

(D) F remains unchanged and R decreases

SPACE FOR ROUGH WORK

SECTION - I(ii) : (Maximum Marks: 24)

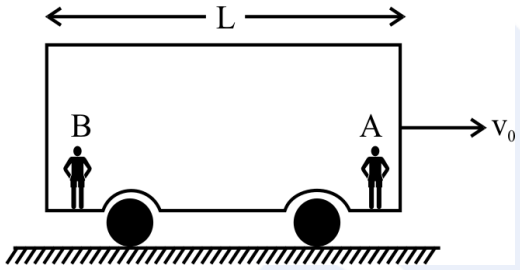
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Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

SPACE FOR ROUGH WORK

7. Two boys standing inside a train compartment throw light ball towards each other inside train. Balls attained same maximum height inside train. Separation between boys is L. Balls have been thrown simultaneously and received simultaneously by A and B. Trains is moving with constant velocity v_0. Mark the correct statement(s) :



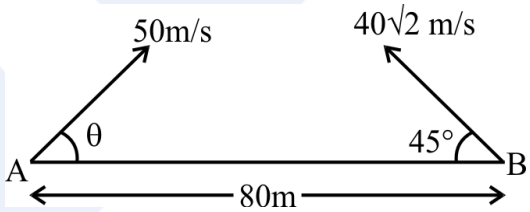
(A) Velocity of projection relative to train is same for both boys.

(B) Velocity of projection relative to train is more for A

(C) Velocity of projection for a ground observer is more for B

(D) Range is more for B for ground observer.

8. Two particles are projected from position 'A' and 'B' seperated by a distance of 80 m on horizontal ground. If both the particles collides in mid air. Select the correct statement [Both particles are projected simultaneously] ($g = 10m/s^2$)



(A) They will collide if angle of projection for 'A' is 37°

(B) Particles will collide in air

(C) Particles will collide at t = 1 sec after projection

(D) They will collide if angle of projection for 'A' is 53°

SPACE FOR ROUGH WORK

9. Two trains one of length $\ell_1 = 630$ m and other of length $\ell_2 = 120$ m move uniformly in two parallel paths in opposite direction with speed $u_1 = 48$ km/h and $u_2 = 102$ km/h respectively.

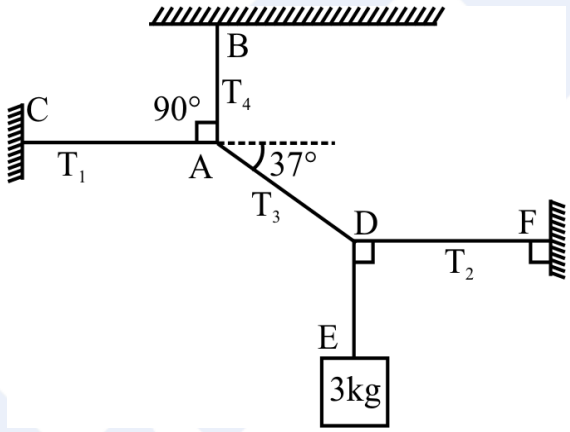
(A) The relative speed of trains is 54 km/hr.

(B) The relative speed of trains is 150 km/hr.

(C) Time taken by one train to completely pass another is 15 s.

(D) Time taken by one train to completely pass another is 18 s.

10. In the figure shown string CA & DF are horizontal & T_1, T_2, T_3 and T_4 are tension in the string CA, DF, AD and AB. then ($g = 10 m/s^2$)



(A) Value of $T_1 = 60$ N, $T_2 = 80$ N

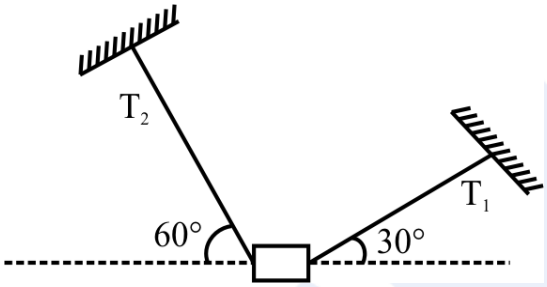
(B) Value of $T_1 = T_2 = 40$ N

(C) Value of $T_3 = 50$ N

(D) Value of $T_4 = 40$ N

SPACE FOR ROUGH WORK

11. A body of mass 5 kg is suspended by strings making angles 60° and 30° with the horizontal as shown in figure. If body is in equilibrium then :- ($g = 10 m/s^2$)



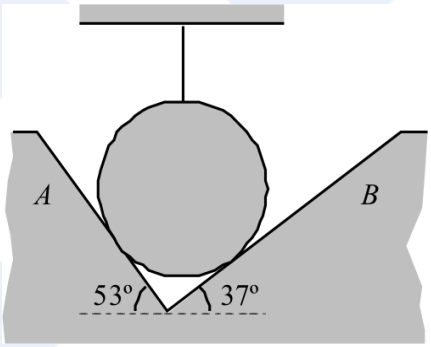
(A) $T_1 = 25$ N

(B) $T_2 = 25$ N

(C) $T_1 = 25\sqrt{3}$N

(D) $T_2 = 25\sqrt{3}$N

12. In the system is shown a 10 kg horizontal cylinder suspended from string from the ceiling and supported by two fixed frictionless slopes A and B. If normal reaction applied by slope A on the cylinder is 30 N, which of the following conclusion can you make? [$g = 10\ m/s^2$]



(A) No tension force is applied by the string on the cylinder.

(B) Tension force applied by the string on the cylinder is 50 N.

(C) Normal reaction applied by the slope B on the cylinder is 60 N.

(D) Normal reaction applied by the slope B on the cylinder is 40 N.

SPACE FOR ROUGH WORK

SECTION-II : (Maximum Marks: 18)

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1. A parrot (popat) is flying horizontally with speed $20\sqrt{2}$ m/s at a height of 2m from ground surface. There is a grain at point B (Jawari grain). As popat passes vertically above of point B, it decided to reach the point B. The maximum possible acceleration of popat is $400\sqrt{3}$ m/s^2. Neglect the reaction time of popat. Then find minimum time (in second) in which popat can reach the point B.



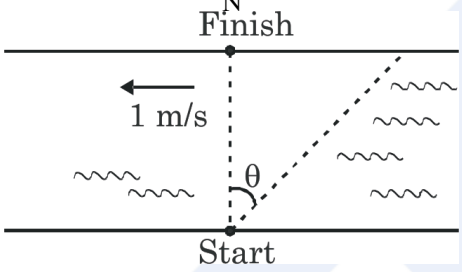


2. At an instant lift is moving with 10 m/s upward and with acceleration 5 m/s^2 upward. At the same instant, a coin is dropped from 7.5m above the floor of lift, find the distance (in m) travelled by coin in ground frame before it strikes the floor :-

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SPACE FOR ROUGH WORK

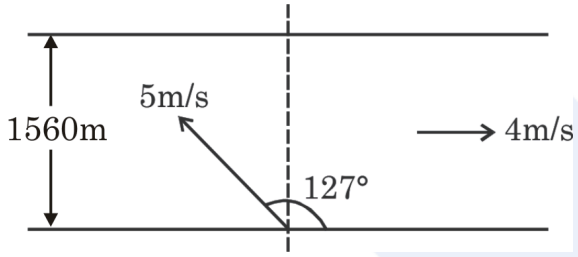
3. A girl wishes to swim across a river to a point directly opposite as shown. She can swim at 2m/s in still water and the river is flowing at 1 m/s. The angle θ (in radian) with respect to the line joining the starting and finishing points at which she should swim is given by $\frac{\pi}{N}$. Fill the value of N.



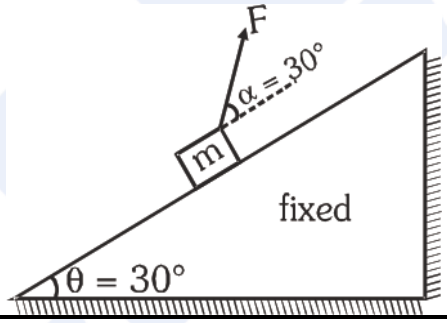
4. If rain drops are falling with speed 12 m/s at angle of 30° with vertically for the observer at rest. Then if observer starts moving with constant acceleration 4 m/s^2 along horizontal. After time T observer observes rain drop again at an angle of 30° with vertical. Then find T.

SPACE FOR ROUGH WORK

5. In the given figure river water is flowing at 4 m/s. A boat is steered at angle 127° from the direction of flow of river. Width of river is 1560 m, and velocity of boat in still water is 5 m/s. Find drift of boat in river (in km).



6. On mass 'm' (as shown in figure) a force 'F' is applied making an angle ' α ' with inclined plane. The value of force is such that it remains stationary with respect to inclined plane.
Find the normal reaction on it (in N). There is no friction between m and inclined plane (given m = 100 √3 gm)



SPACE FOR ROUGH WORK

PART-2 : CHEMISTRY

SECTION - I(i) : (Maximum Marks: 18)

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1. **Assertion (A) :** Among the two O-H bonds in H_2O molecule, the energy required to break the first O–H bond and the other O–H bond is the same.
Assertion (B) : This is because the electronic environment around oxygen is the same even after breakage of one O–H bond is the same.

(A) A and R both are correct, and R is correct explanation of A.

(B) A and R both are correct, but R is not the correct explanation of A.

(C) A is true but R is false.

(D) A and R both are false.

2. There is no S–S bond in:

(A) $S_2O_4^{2-}$ (B) $S_2O_5^{2-}$

(C) $S_2O_3^{2-}$ (D) $S_2O_7^{2-}$

SPACE FOR ROUGH WORK

3. The bond order of N_2 is equal to that of :

(A) O_2 (B) O_2^{2-}

(C) O_2^{2+} (D) None

4. Which of the following has hydrogen bond ?

(A) HF (B) HCl

(C) HBr (D) HI

5. The silicate anion in the mineral is a chain of alternet three $SiO_4^{4\ominus}$ tetrahedral, that share corners with adjacent tetrahedra. The charge of the silicate anion is

(A) –4 (B) –8

(C) –6 (D) –2

6. Which of the following molecule exhibit intermolecular hydrogen bonding :

(A) HBr (liquid) (B) Orthoboric acid

(C) Orthonitrophenol (D) None

SPACE FOR ROUGH WORK

SECTION - I(ii) : (Maximum Marks: 24)

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SPACE FOR ROUGH WORK

7. Which of the following molecule(s) having polar bond :

(A) SO_3 (B) HF

(C) SF_4 (D) XeF_2

8. In which of the following moleucle (s) /ion (s), dz^2 orbital of the central atom involved in hybridisation :

(A) XeF_2 (B) XeF_4

(C) IF_5 (D) PO_4^{3-}

9. Which of the following molecule(s) does not exist :

(A) PCl_5 (B) NCl_5

(C) OF_4 (D) IF_6

SPACE FOR ROUGH WORK

10. Which of the following statement(s) is/are correct regarding ionic compounds?

(A) They are good condutors of electricity at room temperature in solid state

(B) They are more soluble in polar solvent than in non-polar ones

(C) They consist of ions

(D) They generally have high melting and boiling points

11. Which of the following is(are) linear?

(A) I_3^- (B) CO_2

(C) SF_2 (D) XeF_2

12. Which of the following statement is/are incorrect?

(A) sp–hybrid orbitals are at 90° to one another.

(B) sp^3d^2–hybrid orbitals are at 120° to one another.

(C) sp^3d–hybrid orbitals are directed towards the corners of a regular tetrahedron.

(D) sp^3d^2–hybrid orbitals are directed towards the corners of a regular octahedron.

SPACE FOR ROUGH WORK

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1. The bond order of C—O bond in CO_3^{2-} is

2. Total number of species which are planar and polar.
XeF_4, XeF_2, XeF_3^+, SO_2, H_2O

3. No. of π-bonds (pi-bonds) in $C_2(CN)_4$ is.

4. Number of molecules having hybridisation sp^3d are :
(1) IF_7 (2) XeF_2
(3) $I_3^{\ominus}$ (4) SF_4
(5) XeO_3F_2 (6) SO_3

5. What is the % ionic character in HI ? Given electronegativity of H = 2.1 and I = 2.5.
% ionic = $(16 \times \Delta \text{ EN}) + 3.5 \times (\Delta \text{ EN})^2$

6. Total number of species which use d-orbital in their hybridygestion
PCl_5, XeF_2, SF_6, IF_7, SO_3

SPACE FOR ROUGH WORK

PART-3 : MATHEMATICS

SECTION - I(i) : (Maximum Marks: 18)

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1. If five geometric means are inserted between $\frac{3}{2}$ & 486, then product of these geometric means is -

(A) 12^3 (B) 3^{12}

(C) 15^3 (D) 3^{15}

2. Which of the following can't be n^{th} term of an A.P.

(A) $t_n = \frac{n^2-4}{n+2}$ (B) $t_n = \sqrt{4n^2+4n+1}$

(C) $t_n = n^2 - (n+2)^2$ (D) $t_n = n^2 - (2n+1)^2$

SPACE FOR ROUGH WORK

3. In a convex polygon, all the interior angles are in A.P. If smallest angle is 159° and the largest angle is 177°, then the number of sides in the polygon, is -

(A) 21 (B) 27

(C) 30 (D) 31

4. If $x = \frac{4}{3} - \frac{4x}{9} + \frac{4x^2}{27} - \ldots\ldots\ldots \infty$, then x is equal to –

(A) only 1 (B) 1 or –4

(C) only –4 (D) –1 or 4

SPACE FOR ROUGH WORK

5. Let S_n denotes the sum of first n terms of an arithmetic progression whose first term is -4 and common difference is 1. If $V_n = 2S_{n+2} - 2S_{n+1} + S_n$ $(n \in N)$, then

(A) $V_n = \frac{-9n^2 + 5n - 12}{2}, n \in N$

(B) $V_n = \frac{n^2 + 5n + 12}{2}, n \in N$

(C) Minimum value of V_n is -9

(D) Minimum value of V_n is $-\frac{73}{8}$

6. If $(1.5)^{30} = K$, then the value of $\sum_{n=2}^{29} (1.5)^n$ is -

(A) $2K - 3$

(B) $K + 1$

(C) $2K + 7$

(D) $2K - \frac{9}{2}$

SPACE FOR ROUGH WORK

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- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

SPACE FOR ROUGH WORK

7. If x,y,z > 0 and x + y + z = 9, then correct statement is (are) -

(A) Maximum value of xyz is 27

(B) Maximum value of xy^2z is $\frac{3^8}{2^6}$

(C) Maximum value of x^2yz^3 is $\frac{3^8}{2^4}$

(D) Maximum $x^2y^3z^4$ is $2^{10}.3^3$

8. Let A,G,H are arithmetic, geometric and harmonic mean between two given positive numbers x and y. Then -

(A) A^2, AG, AH are in G.P.

(B) Quadratic $Ax^2 + Gx + H = 0$ has imaginary roots

(C) Quadratic $Ax^2 + Gx + H = 0$ has real roots

(D) $A \geq G \geq H$

9. If a, b, c are in A.P., b, c, d are in G.P. & c, d, e are in H.P., then –

(A) a, e, c are in G.P.

(B) a, c, e are in G.P.

(C) $\frac{c+a}{c-a} = \frac{e+c}{e-c}$

(D) $\frac{e+c}{e-c} = \frac{a+e}{(a-e)}$

SPACE FOR ROUGH WORK

10. If α and β are the roots of the equation $ax^2 + bx + c = 0$ and $\alpha\beta = 3$, where a, b, c are in arithmetic progression, then $(\alpha^2 + \beta^2)$ is equal to-

(A) 2

(B) -2

(C) $\alpha^3 + \beta^3$

(D) $\alpha + \beta$

11. If n^{th} term of an arithmetic progression is $T_n = (k - 3)n^2 + (k + 1)n - 3$, then –

(A) $k = 3$

(B) common difference = 8

(C) common difference = 4

(D) first term = 1

12. If A_i, G_i, H_i $(i = 1, 2, 3)$ are respectively three A.M's, G.M's and H.M's between two positive numbers, then -

(A) $H_1A_3 = G_1G_3$

(B) $\frac{A_1}{A_3} = \frac{H_1}{H_3}$

(C) $A_1H_3 = G_1G_3$

(D) $A_2H_2 = G_2^2$

SPACE FOR ROUGH WORK

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

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1. If the sum of series $\frac{1}{1.(1+d)} + \frac{1}{(1+d)(1+2d)} + \ldots + \frac{1}{(1+9d)(1+10d)}$ is equal to 5, then d is equal to :

2. If the sum $\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \ldots$ upto 20 terms is $\frac{k}{21}$, then $\frac{k}{100}$ is

3. If $1(2010) + 2(2009) + 3(2008) + ... + 2010(1) = (335)(2011)\lambda$, then $\frac{\lambda}{506}$ is

4. If $a_1, a_2, ..., a_8$ are in geometric progression. Their sum is 24 and the sum of their reciprocals is 8. The value of their product is

5. Let sum of first 'n' terms of two AP's are $\frac{n}{2}(7n+13)$ and $\frac{n}{2}(313-13n)$ respectively. If their corresponding n^{th} terms are equal, then the value of 'n' is

6. If $x = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots \infty$

$y = \frac{1}{3} - \frac{1}{3^2} + \frac{1}{3^3} - \frac{1}{3^4} + \ldots \infty$

$z = \frac{1}{2.3} + \frac{1}{2^2.3^2} + \frac{1}{2^3.3^3} + \ldots \infty$

then value of $\frac{xy}{z}$ is

SPACE FOR ROUGH WORK

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